

PARTHA TARAFDAR

Chicago, IL | 630-486-2480 | tarafdarpartha2025@gmail.com | parthatarafdar.com | GitHub | LinkedIn

Software engineer and MS in Artificial Intelligence candidate who builds AI systems end to end — retrieval, evaluation, and deployment. Founder of Polaris AI Technologies LLC, with two live products shipped: Janus, a grounded question-answering engine, and CareerLaunch, a job-application tracker.

EDUCATION

MS in Artificial Intelligence, DePaul University

Expected 2027

- Current GPA: 3.6

BS in Computer Science, University of Wisconsin–Madison

May 2025

- Coursework: Algorithms, Artificial Intelligence, Data Science, Big Data Systems, Software Engineering, Human-Computer Interaction, Statistics.
- Honors: Dean's List 2020, 2021.

WORK EXPERIENCE

Founder / Full-Stack Developer — Polaris AI Technologies LLC

June 2025 – Present

Chicago, IL

- Built and deployed Janus (leagueoflegends.polaris-ai.org), a retrieval-augmented question-answering engine that answers from real game data, cites every claim, and refuses when the data does not cover the question — implemented without a RAG framework.
- Designed three retrieval strategies behind a single intent router: a version-pinned prose corpus in Postgres/pgvector, typed SQL tables for exact numeric lookups, and a sanctioned live third-party MCP API for daily-changing statistics.
- Built an evaluation harness with a hand-curated, double-reviewed 40-question golden set; scored 5.00/5 faithfulness under an independent judge model and promoted retrieval changes only on measured evidence (corpus-composition filtering improved hit@1 from 66.7% to 74.5%).
- Benchmarked five embedding models against the golden set after auditing and overturning an earlier flawed result; provisioned and deployed the full stack on a Linux VPS — Docker Compose services (FastAPI, Postgres/pgvector, React, nginx) behind Caddy with automatic HTTPS on a custom domain.
- Founded, designed, built, and launched CareerLaunchOnline.com, a job-application tracker for statuses, resume versions, and weekly/monthly analytics — React (Vite), Flask REST API, PostgreSQL, Firebase Auth (Google SSO).

Website Administrator (Summer Volunteer) — Badminton Warehouse

May–Aug 2023, 2024

Schaumburg, IL

- Customized and maintained a Shopify e-commerce site using Liquid, HTML/CSS, and JavaScript, improving layout consistency and mobile responsiveness across product pages.
- Developed Python scripts to automate bulk product updates and keyword performance tracking, reducing manual update time by over 60%.
- Optimized product metadata and descriptions for SEO, achieving top-3 Google rankings for most target categories; implemented analytics to monitor traffic and conversion behavior.

Software Developer Summer Intern — US Cellular

May – Aug 2022

Schaumburg, IL

- Automated monthly operations reporting by building a Flask and SQLAlchemy API with optimized SQL queries, reducing runtime by 33% and saving hours of manual effort.
- Developed a React internal dashboard to visualize report data, making reports accessible to over 40 users.
- Collaborated on an Agile team of four, applying iterative development and testing to incrementally deliver features.

SELECTED PROJECTS

Distributed Weather Station — Python, Apache Cassandra, PySpark, gRPC, HDFS

- Built a gRPC backend managing Wisconsin weather-station data in Cassandra, with PySpark parsing and filtering, and fault-tolerant ingestion that stays correct when nodes drop out.

Persona Chatbot — React, OpenAI API, Bootstrap, Local Storage

- Built a persona-driven chatbot on the GPT-4o API with dynamic persona switching, real-time streaming UI, and persistent session memory.

Voice Assistant — React, Wit.ai, REST API, NLP

- Built a natural-language voice chatbot with custom-trained intents for help, chatroom listing, and message retrieval.

Chat Mobile App — React Native, Expo, JWT, Secure Store

- Built a secure mobile chat app with login, registration, chatroom navigation, and guest mode; expo-secure-store for JWT handling.

SKILLS & INTERESTS

Languages: Python, JavaScript, Java, C++, C, SQL, HTML/CSS.

Frameworks & Tools: React, FastAPI, Flask, SQLAlchemy, PostgreSQL, pgvector, Docker, Caddy, nginx, Spark, Cassandra, Kafka, Git.

AI & ML: Retrieval-augmented generation, embeddings and vector search, hybrid retrieval and reranking, LLM evaluation (golden sets, LLM-as-judge, faithfulness), prompt design, model APIs (OpenAI, OpenRouter), MCP.